

EMPLOYEE PERFORMANCE MAPPING

Course-End Project 1 - SQL

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Role: Business Analyst

Case Organization: ScienceQtech

A structured SQL solution for employee performance, salary, experience, geography and HR appraisal analysis.

1. Executive Summary

This project applies SQL fundamentals to employee performance mapping for ScienceQtech. It supports the HR Analysis Department during the annual appraisal cycle through employee, performance, salary, role, experience, location and project reporting.

2. Business Scenario

ScienceQtech is a data-science startup with projects including fraud detection, market basket analysis, self-driving cars, supply-chain optimization, early lung-cancer detection, customer sentiment analysis and drug discovery.

3. Objectives

Create and validate an employee database; retrieve employee details; classify performance ratings; identify leadership roles; compare departmental ratings; analyze salaries; rank experience; create views; optimize search through indexing; and analyze geographic salary distribution.

4. Scope

The project covers the three supplied datasets and the SQL analyses required by the problem statement. Payroll, attendance, recruitment, leave management and predictive HR models are outside the stated scope.

5. Input Data Model

Dataset	Key content
emp_record_table	Employee identity, gender, role, department, experience, geography, salary, rating, manager and project mapping.
proj_table	Project identity/name, domain, dates, development quarter and status.
data_science_team	Employee identity, gender, role, department, experience and geography for the Data Science team.

6. Database Creation

```
CREATE DATABASE employee;
USE employee;
SHOW TABLES;
DESCRIBE emp_record_table;
SELECT * FROM emp_record_table LIMIT 10;
```

Business interpretation: Creates and validates the database.

Insert screenshot of the actual executed SQL output in the final LMS submission.

7. Employee and Department Details

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT
FROM emp_record_table;
```

Business interpretation: Creates an HR employee/department directory.

Insert screenshot of the actual executed SQL output in the final LMS submission.

8. Performance Rating Categories

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT, EMP_RATING FROM emp_record_table WHERE EMP_RATING < 2;

SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT, EMP_RATING FROM emp_record_table WHERE EMP_RATING BETWEEN 2
AND 4;

SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT, EMP_RATING FROM emp_record_table WHERE EMP_RATING > 4;
```

Business interpretation: Segments employees into the three requested rating groups.

Insert screenshot of the actual executed SQL output in the final LMS submission.

9. Leadership Positions

```
SELECT EMP_ID, FIRST_NAME, ROLE, DEPT FROM emp_record_table
WHERE ROLE LIKE '%MANAGER%' OR ROLE LIKE '%PRESIDENT%' OR ROLE LIKE '%CEO%';
```

Business interpretation: Identifies leadership roles.

Insert screenshot of the actual executed SQL output in the final LMS submission.

10. Department Maximum Rating

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPT, EMP_RATING,
MAX(EMP_RATING) OVER (PARTITION BY DEPT) AS MAX_DEPT_RATING
FROM emp_record_table ORDER BY DEPT, EMP_RATING DESC;
```

Business interpretation: Shows individual rating with departmental maximum.

Insert screenshot of the actual executed SQL output in the final LMS submission.

11. Salary Range by Role

```
SELECT ROLE, MIN(SALARY) AS MIN_SALARY, MAX(SALARY) AS MAX_SALARY
FROM emp_record_table GROUP BY ROLE ORDER BY ROLE;
```

Business interpretation: Analyzes role-level salary ranges.

Insert screenshot of the actual executed SQL output in the final LMS submission.

12. Experience Ranking

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, EXP,
RANK() OVER (ORDER BY EXP DESC) AS EXPERIENCE_RANK
FROM emp_record_table;
```

Business interpretation: Ranks employees by experience.

Insert screenshot of the actual executed SQL output in the final LMS submission.

13. Salary View Above 6000

```
CREATE VIEW employee_salary_above_6000 AS
SELECT EMP_ID, FIRST_NAME, LAST_NAME, COUNTRY, CONTINENT, ROLE, SALARY
FROM emp_record_table WHERE SALARY > 6000;
SELECT * FROM employee_salary_above_6000;
```

Business interpretation: Creates a reusable salary-threshold view.

Insert screenshot of the actual executed SQL output in the final LMS submission.

14. Country-wise Salary View

```
SELECT COUNTRY, EMP_ID, FIRST_NAME, LAST_NAME, ROLE, SALARY
FROM employee_salary_above_6000 ORDER BY COUNTRY, SALARY DESC;
```

Business interpretation: Organizes qualifying employees by country.

Insert screenshot of the actual executed SQL output in the final LMS submission.

15. Index and Execution Plan

```
EXPLAIN SELECT * FROM emp_record_table WHERE FIRST_NAME='Eric';
CREATE INDEX idx_emp_first_name ON emp_record_table(FIRST_NAME);
EXPLAIN SELECT * FROM emp_record_table WHERE FIRST_NAME='Eric';
```

Business interpretation: Demonstrates indexing and execution-plan comparison.

Insert screenshot of the actual executed SQL output in the final LMS submission.

16. Average Salary by Geography

```
SELECT CONTINENT, COUNTRY, ROUND(AVG(SALARY),2) AS AVG_SALARY
FROM emp_record_table GROUP BY CONTINENT, COUNTRY ORDER BY CONTINENT, COUNTRY;
```

Business interpretation: Calculates average salary by continent and country.

Insert screenshot of the actual executed SQL output in the final LMS submission.

17. Maximum Salary

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPT, SALARY
FROM emp_record_table WHERE SALARY=(SELECT MAX(SALARY) FROM emp_record_table);
```

Business interpretation: Uses a subquery to identify maximum-salary employee(s).

Insert screenshot of the actual executed SQL output in the final LMS submission.

18. Training Review

```
SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPT, EMP_RATING
FROM emp_record_table WHERE EMP_RATING < 2 ORDER BY EMP_RATING ASC;
```

Business interpretation: Creates a lower-rating list for further HR skill/training review.

Insert screenshot of the actual executed SQL output in the final LMS submission.

19. Job Role Standardization

```
SELECT DISTINCT ROLE FROM emp_record_table ORDER BY ROLE;
```

Business interpretation: Creates a unique role list for comparison with the approved job-profile master.

Insert screenshot of the actual executed SQL output in the final LMS submission.

20. Bonus Analysis Note

The overview mentions bonus calculation, but no approved bonus percentage or formula is provided in the supplied problem statement. An exact bonus has therefore not been invented. Once supplied, Bonus = Salary x Approved Bonus Rate.

21. Business Analyst Perspective

Business Requirement -> Data Requirement -> SQL Logic -> Query Result -> Business Interpretation -> HR Decision Support.

22. Benefits

Consistent HR reporting, performance visibility, compensation analysis, experience analysis, reusable views and improved retrieval performance.

23. Limitations

Ratings and salaries require organizational context. Final appraisal, training, promotion and compensation decisions should also consider role requirements, experience, manager feedback, location and approved policies.

24. Future Enhancements

Appraisal history, skill matrices, training/certification tracking, salary bands, job-profile masters, project-performance measures, attrition analysis and Power BI dashboards.

25. Conclusion

The project demonstrates practical SQL use for HR analysis through filtering, aggregation, window functions, ranking, views, indexing, execution plans and subqueries.

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